

Chapter 1

MathStat Preliminaries

1.1

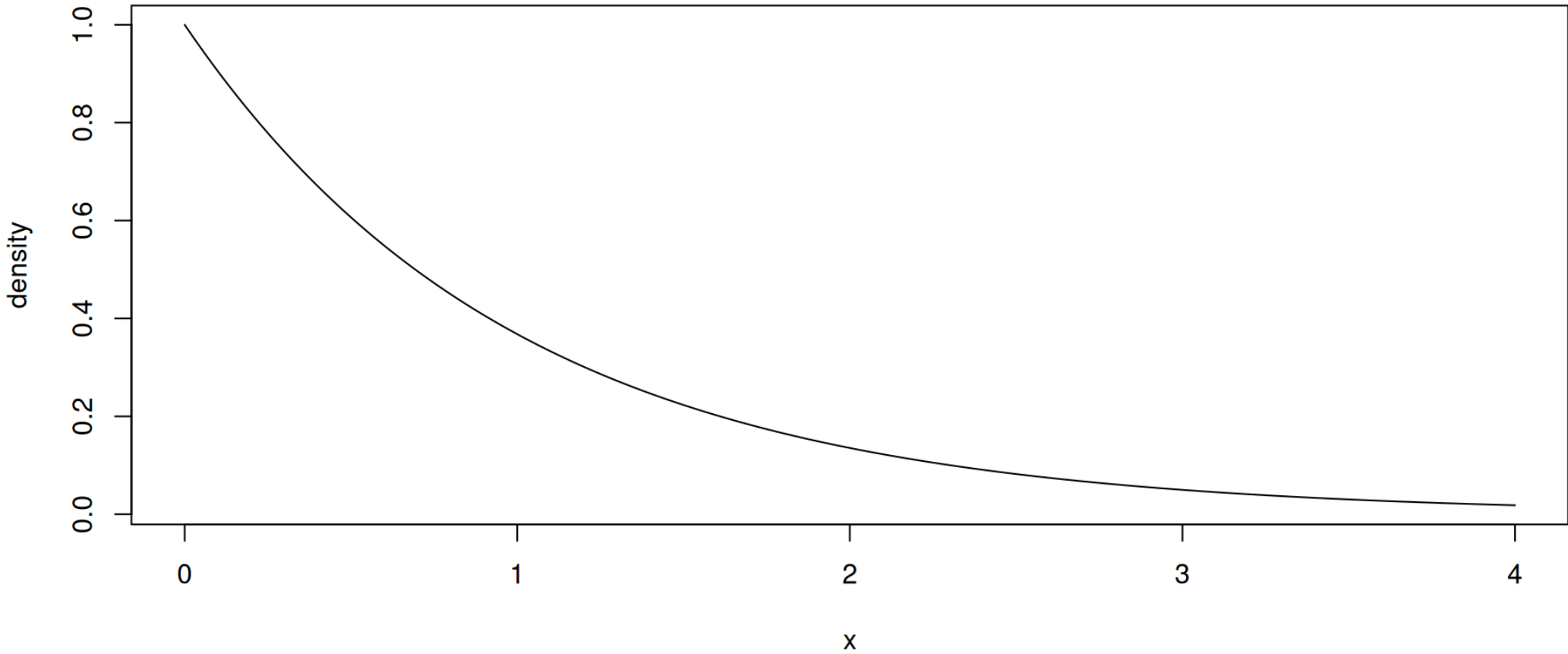
Wait. Where are we going?

Exponential pdf

If $X \sim \text{Exponential}(\lambda)$, then X has the pdf

$$f(x) = \lambda \exp(-\lambda x) I_{[0, \infty]}(x).$$

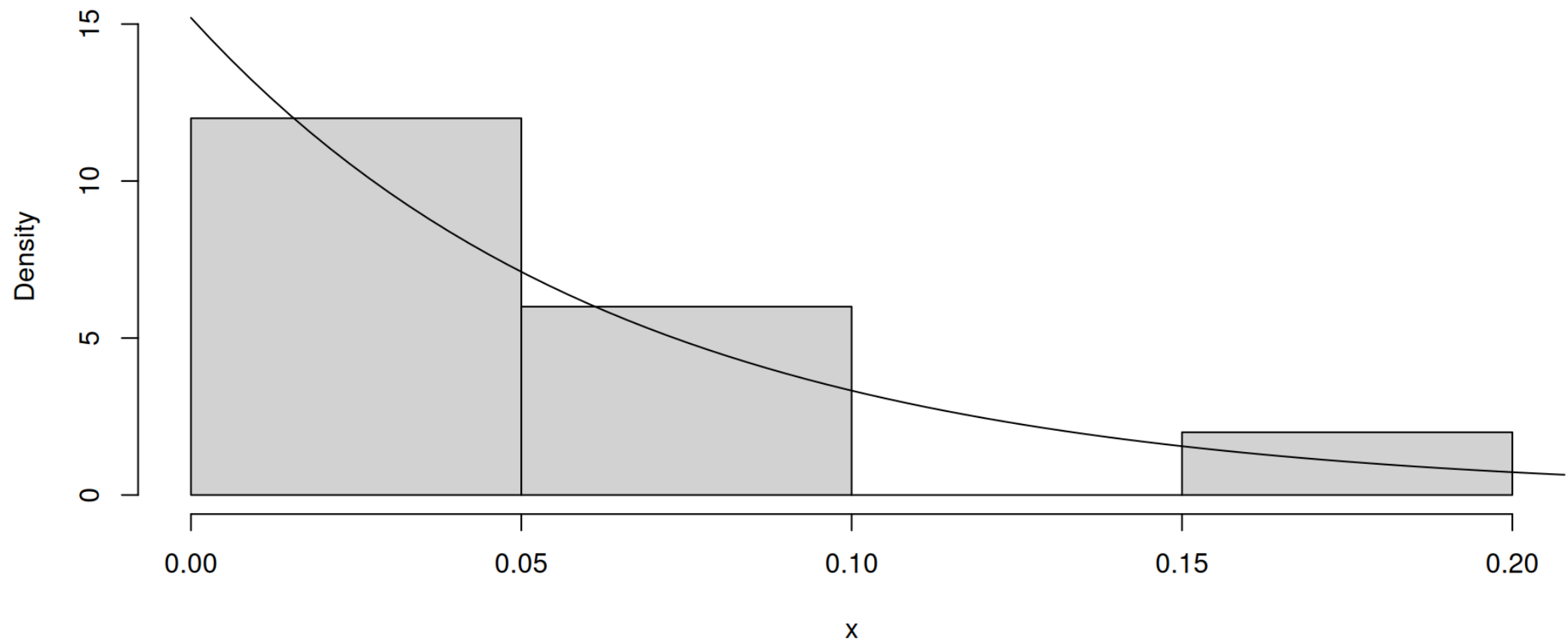
Exponential pdf visualized



Sample

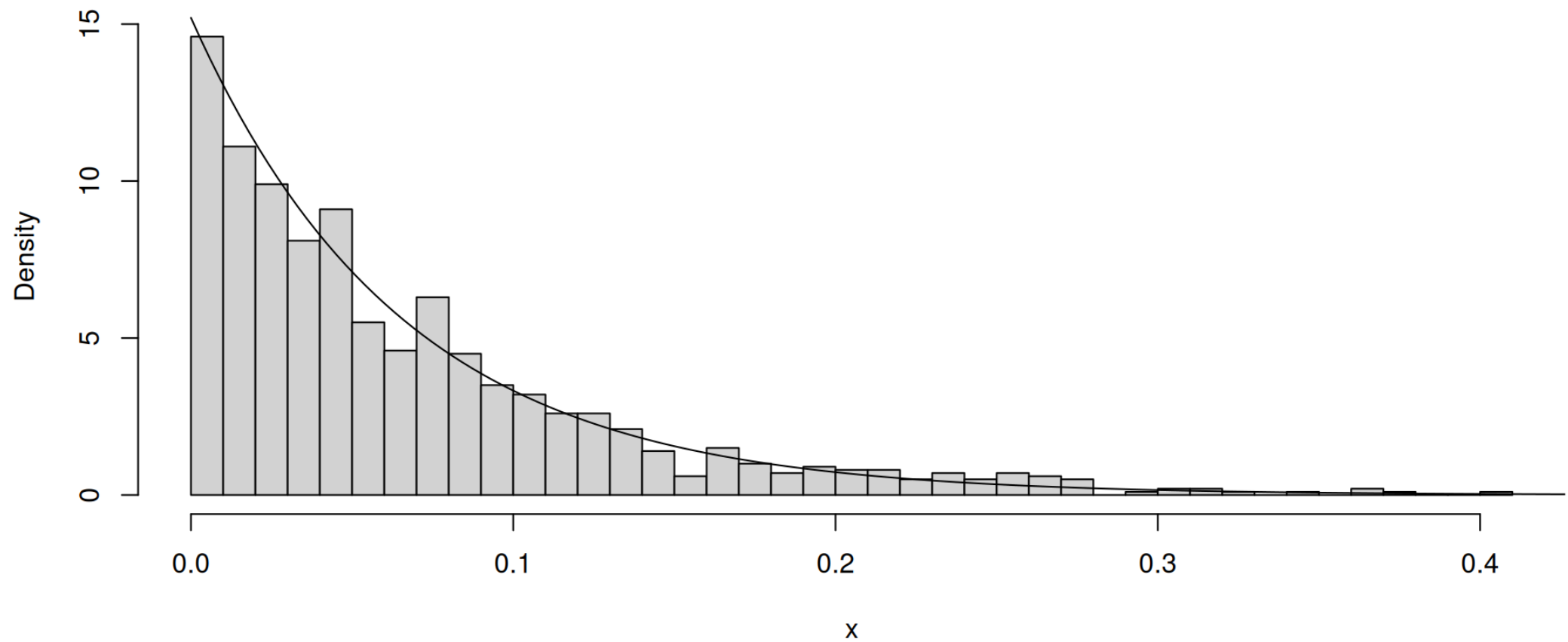
A sample of n realization of X , denoted $X_1, X_2, \dots, X_n$, can be used to approximate the distribution of X .

Histogram for 10 realizations



A histogram of 10 realizations from an Exponential(15.2).

Histogram for 1000 realizations



A histogram of 1000 realizations from an Exponential(15.2).

1.1.1

A very special trick

Approach

We can avoid doing actual integration if we can manipulate the integrand to look more like a pdf, which must integrate to 1 of the range of the random variable.

Example: Determine

$$\int_0^{\infty} 3 \exp(-2x) dx.$$

Special trick (cont)

1.2

Transformations of Random Variables

1.2.1

The discrete case and the binomial distribution

The binomial pmf

$X \sim \text{Binomial}(n, p)$ has the pmf

$$f(x) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} I_{\{0,1,\dots,n\}}(x).$$

Determine the pmf of $Y := n - X$.

1.2.2

The Continuous Case and the Gamma Distribution

Continuous Transformation PDF

Let X be a continuous random variable with pdf $f_X(x)$. Let $Y = g(X)$, where g is invertible and differentiable. Then the pdf for Y is

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Proof

Proof (cont)

Proof (cont)

Proof (cont)

Example 1.2.2

A continuous random variable $X \sim \text{Gamma}(\alpha, \beta)$ has pdf

$$f_X(x) = \frac{1}{\Gamma(\alpha)} \beta^\alpha x^{\alpha-1} \exp(-\beta x) I_{(0,\infty)}(x).$$

Example 1.2.2 (cont)

Let $Y := cX$ with $c > 0$. Determine the pdf of Y .

Example 1.2.2 (cont)

The shape parameter

The α term is the “shape” parameter of the Gamma distribution.

The kernel (non-constant) part of the pdf of the Gamma distribution is

$$x^{\alpha-1} \exp(-\beta x) I_{(0,\infty)}(x).$$

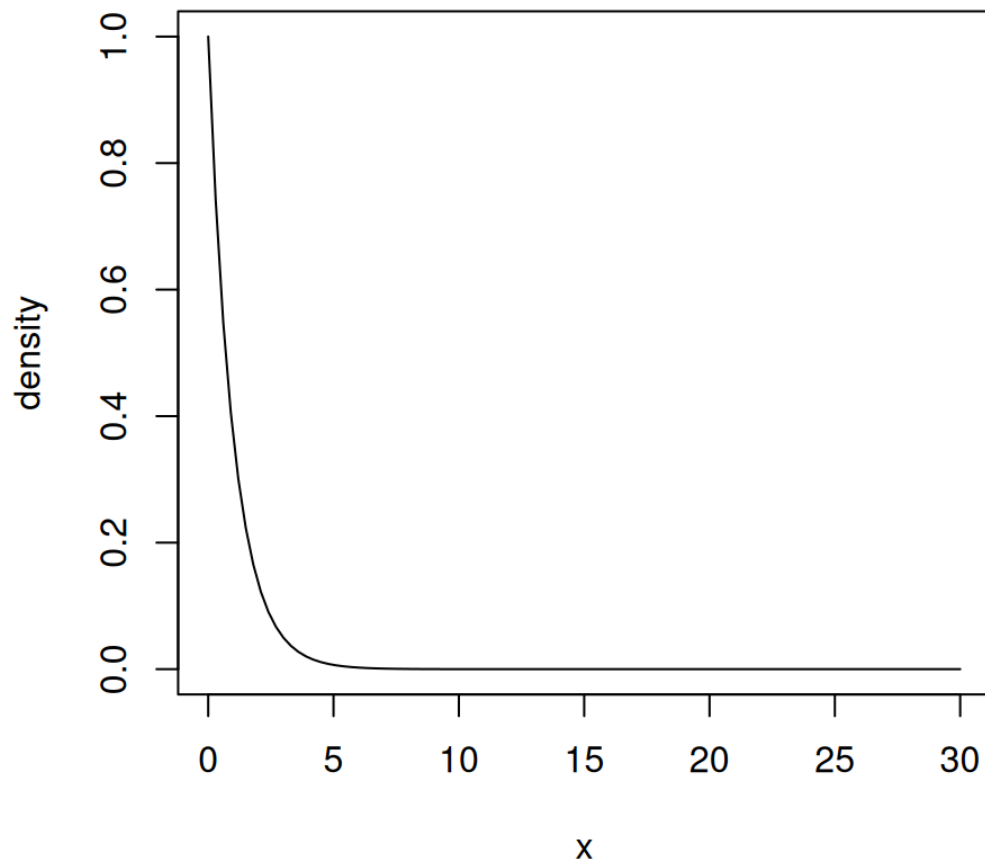
The shape parameter

The exponential function part is an unscaled exponential density.

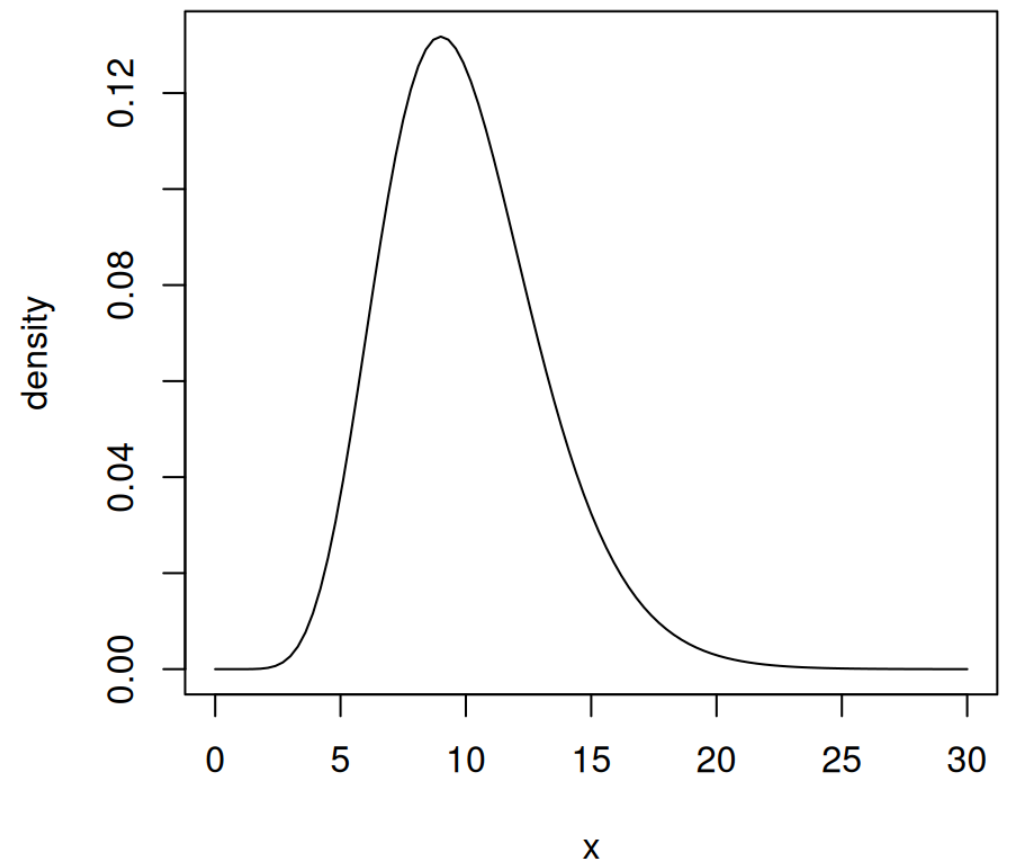
- $\exp(-\beta x)$ dominates $x^{\alpha-1}$ for large x .
- For small x , the $x^{\alpha-1}$ term controls the shape.

Shape parameter visualized

$\alpha=1, \beta=1$



$\alpha=10, \beta=1$



The α terms controls the shape of the Gamma distribution.

The Gamma Function

The pdf for the gamma distribution is defined using the **gamma function**, denoted by $\Gamma(\alpha)$.

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} \exp(-x) dx.$$

The Gamma Function (cont)

The Gamma Function (cont)

The Gamma Function (cont)

1.3

Bivariate Transformations

Bivariate Transformation PDF

Suppose the X_1 and X_2 are jointly continuous random variables with pdf $f_{X_1, X_2}(x_1, x_2)$. Let $Y_1 = g_1(X_1, X_2)$ and $Y_2 = g_2(X_1, X_2)$. For h_1 and h_2 differentiable, let

$$X_1 = h_1(Y_1, Y_2) \text{ and } X_2 = h_2(Y_1, Y_2).$$

Then

$$f_{Y_1, Y_2}(y_1, y_2) = f_{X_1, X_2}(h_1(y_1, y_2), h_2(y_1, y_2)) |J|,$$

where $|J|$ is the absolute value of the Jacobian.

Bivariate transformation pdf continued

The absolute value of the Jacobian is

$$J = \det \begin{bmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{bmatrix}.$$

Example 1.3.1

Let $X_1, X_2 \stackrel{i.i.d.}{\sim} \text{Gamma}(\alpha, \beta)$

Determine the pdf of

$$Y = \frac{X_1}{X_1 + X_2}.$$

Example 1.3.1 (cont)

Example 1.3.1 (cont)

Example 1.3.1 (cont)

Example 1.3.1 (cont)

The Beta Distribution

The continuous random variable $X \sim \text{Beta}(a, b)$ had pdf

$$f(x) = \frac{1}{\mathcal{B}(a, b)} x^{a-1} (1 - x)^{b-1} I_{(0,1)}(x),$$

for $a, b > 0$, where

$$\mathcal{B}(a, b) = \int_0^1 x^{a-1} (1 - x)^{b-1} dx$$

denotes the **beta function**, which normalizes the kernel of the Beta distribution.

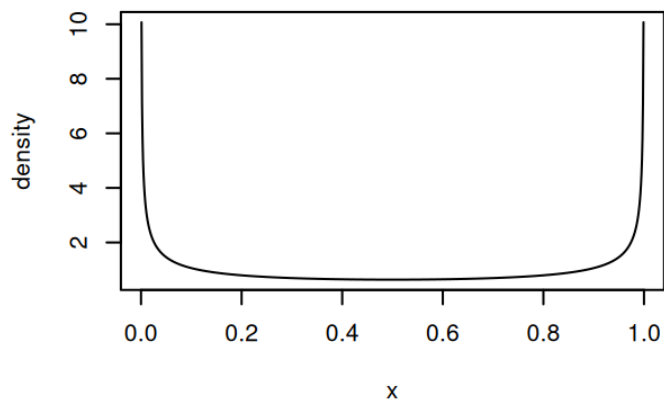
The Beta Distribution (cont)

The Beta distribution is a flexible distribution for modeling a random variable between 0 and 1.

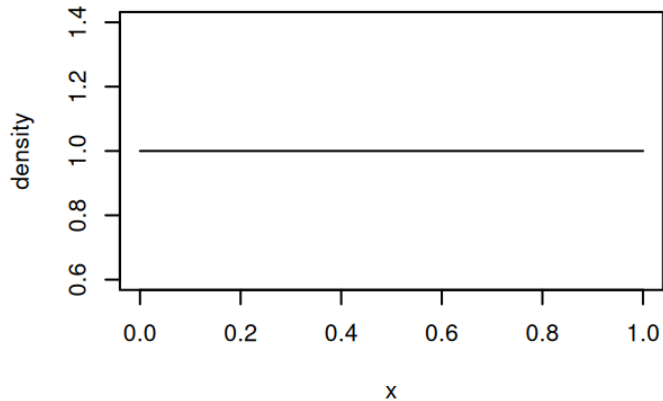
Note: The Uniform(0, 1) distribution is a special case of the Beta distribution with

The Beta Distribution (cont)

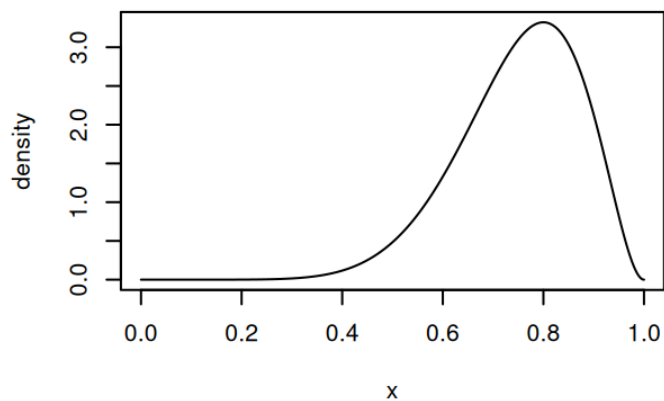
a = 0.5, b = 0.5



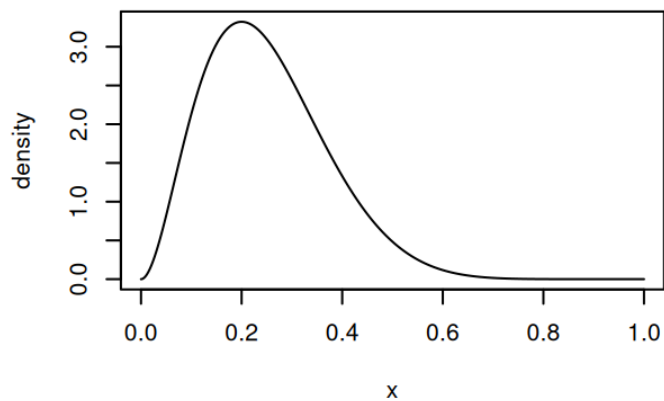
a = 1, b = 1



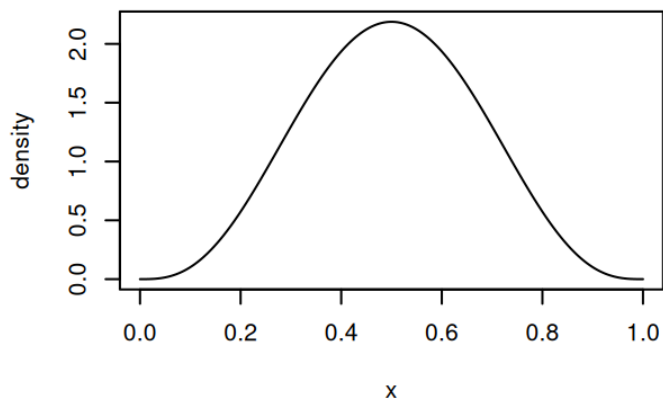
a = 9, b = 3



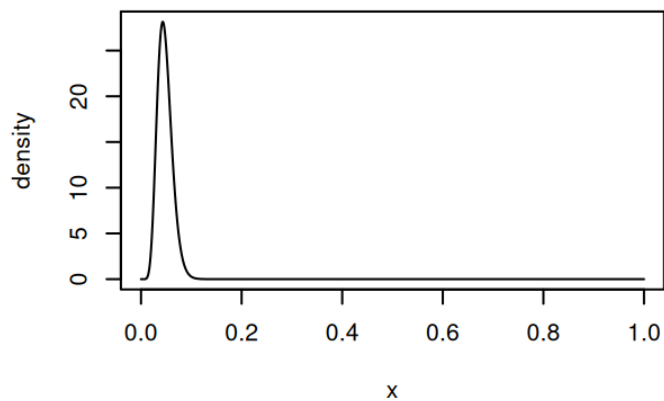
a = 3, b = 9



a = 4, b = 4



a = 10, b = 200



The Beta Function

$$\mathcal{B}(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

Proof:

The Beta Function (cont)

1.4

Minimums and Maximums

Order statistics notation

Let $X_1, X_2, \dots, X_n$ denote a sample of observations. The order statistics are denoted as:

- $X_{(1)} = \min(X_1, X_2, \dots, X_n)$.
- $X_{(2)} =$ 2nd smallest of $X_1, X_2, \dots, X_n$.
- $X_{(n)} = \max(X_1, X_2, \dots, X_n)$.

1.4.2

The distribution of a minimum by
example

Example 1.4.1

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Exponential}(\lambda)$. Determine the distribution of $X_{(1)}$.

Example 1.4.1 (cont)

Example 1.4.1 (cont)

Example 1.4.1 (cont)

Example 1.4.1 (cont)

1.4.3

The general distribution of
minimums and maximums

Distribution of the minimum

Let $X_1, X_2, \dots, X_n$ be a random sample from a continuous distribution. Determine the distribution of $X_{(1)}$.

Distribution of the minimum (cont)

Distribution of the minimum (cont)

Distribution of the minimum (cont)

PDF for a minimum

Let $X_1, X_2, \dots, X_n$ be a random sample from a continuous distribution with pdf f and cdf F .

The pdf for $X_{(1)}$ is

$$f_{X_{(1)}} = n[1 - F(x)]^{n-1} f(x).$$

Distribution of the maximum

Let $X_1, X_2, \dots, X_n$ be a random sample from a continuous distribution. Determine the distribution of $X_{(n)}$.

Distribution of the maximum (cont)

Distribution of the maximum (cont)

Distribution of the maximum (cont)

PDF for a maximum

Let $X_1, X_2, \dots, X_n$ be a random sample from a continuous distribution with pdf f and cdf F .

The pdf for $X_{(n)}$ is

$$f_{X_{(n)}} = n[F(x)]^{n-1} f(x).$$

1.5

Moment Generating Functions (mgfs)

Definition of the mgf

The moment generating function (mgf) of a random variable X is

$$M_X(t) = E[e^{tX}].$$

Uses of mgfs

Mgfs are often useful in two settings:

1. Determining the distribution of a transformation of 1 or more random variables (particularly the sum of n i.i.d. random variables.)
2. Determine the “moments” of a random variable, i.e.,
 $E(X), E(X^2), E(X^3), \dots$

1.5.1

The expectation of a function X

Example 1.5.1

Suppose the random variable X has the following distribution.

x	-1	0	1
$f(x)$	4/12	3/12	5/12

Determine $E(X^2)$.

Example 1.5.1 (cont)

Example 1.5.1 (cont)

Law of the Unconscious Statistician

If X is a discrete random variable, then

$$E[g(X)] = \sum_x g(x) f_X(x),$$

where $f_X(x)$ is the pmf of X .

If X is a continuous random variable, then

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx,$$

where $f_X(x)$ is the pdf of X .

Proof (specific case)

Proof (specific case)

Proof (specific case)

Proof (specific case)

1.5.2

Finding mgfs and the Poisson
distribution

The Poisson distribution

A discrete random variable $X \sim \text{Poisson}(\lambda)$ if it has the pmf

$$f_X(x) = \frac{e^{-\lambda} \lambda^x}{x!} I_{\{0,1,2,\dots\}}(x).$$

Poisson interpretation

A Poisson random variable can be thought of as the number of arrivals observed in a fixed amount of time where:

1. The mean time between arrivals is constant.
2. The number of arrivals in non-overlapping time periods are independent.

Example 1.5.2

Determine the mgf for the Poisson distribution

Example 1.5.2 (cont)

Example 1.5.2 (cont)

Example 1.5.3

Suppose that $X \sim \text{Exponential}(\lambda)$. Determine the mgf of X .

Example 1.5.3 (cont)

Example 1.5.3 (cont)

Example 1.5.4

Suppose that $X \sim N(\mu, \sigma^2)$ with pdf

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2\sigma^2}(x - \mu)^2\right].$$

Determine the mgf of X .

Example 1.5.4 (cont)

Example 1.5.4 (cont)

Example 1.5.4 (cont)

1.5.3

Finding moments

Defining moments

The **moments** of a random variable X are expectations of the form $E(X^k)$ for $k = 1, 2, 3, \dots$

Computing moments

The k th moment of a random variable X can be computed as $M_X^{(k)}(0)$, where $M_X^{(k)}(0)$ is the k th derivative of the mgf of X evaluated at zero.

Mgf moment proof

Mgf moment proof (cont)

Mgf moment proof (cont)

Example 1.5.5

Determine the mean and variance of the Poisson(λ) distribution.

Example 1.5.5 (cont)

Example 1.5.5 (cont)

1.5.4

The mgf uniquely determines a distribution

The DNA of the mgf

The mgf of a random variable uniquely determines its distribution.

Example: If we find the mgf of a random variable X is $\exp[15(e^t - 1)]$, then $X \sim \text{Poisson}(15)$ since it matches the mgf of a Poisson random variable with $\lambda = 15$.

1.5.6

Sums of iid random variables

Mgf of sum of independent r.v.s

Let $X_1, X_2, \dots, X_n$ be independent random variables. The mgf of

$$Y := \sum_{i=1}^n X_i$$

is

$$M_Y(t) = \prod_{i=1}^n M_{X_i}(t).$$

Proof

Proof (cont)

Proof (cont)

Mgf of sum of i.i.d. r.v.s

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} F$. The mgf of

$$Y := \sum_{i=1}^n X_i$$

is

$$M_Y(t) = [M_X(t)]^n.$$

Example 1.5.6

Suppose that $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Poisson}(\lambda)$. Determine the distribution of $Y := \sum_{i=1}^n X_i$.

Example 1.5.6 (cont)

Example 1.5.6 (cont)

Example 1.5.7

Suppose that $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Exponential}(\lambda)$. Determine the distribution of $Y := \sum_{i=1}^n X_i$.

Example 1.5.7 (cont)

Example 1.5.7 (cont)

Example 1.5.8

Suppose that $X_1, X_2, \dots, X_n$ be independent random variables with $X_i \sim \text{Poisson}(\lambda_i)$. Determine the distribution of $Y := \sum_{i=1}^n X_i$.

Example 1.5.8 (cont)

Example 1.5.8 (cont)

Example 1.5.8 (cont)

1.6

General Order Statistics

1.6.2

The joint distribution of the
minimum and maximum

Min/Max cdf

For an i.i.d. sample of continuous random variables $X_1, X_2, \dots, X_n$ with cdf F , the joint cdf of $(X_{(1)}, X_{(n)})$ is

$$F_{X_{(1)}, X_{(n)}}(x, y) = [F(y)]^n - [F(y) - F(x)]^n,$$

for $x < y$ and

$$F_{X_{(1)}, X_{(n)}}(x, y) = [F(y)]^n$$

for $x \geq y$.

Proof of min/max cdf

Proof of min/max cdf (cont)

Proof of min/max cdf (cont)

Min/Max pdf

For an i.i.d. sample of continuous random variables $X_1, X_2, \dots, X_n$ with cdf F , the joint pdf of $(X_{(1)}, X_{(n)})$ is

$$f_{X_{(1)}, X_{(n)}}(x, y) = n(n-1)[F(y) - F(x)]^{n-2} f(x)f(y),$$

for $x < y$.

Proof of min/max pdf

Example 1.6.1

Determine the joint pdf of $(X_{(1)}, X_{(15)})$ when

$X_1, X_2, \dots, X_{15} \stackrel{i.i.d.}{\sim} \text{Uniform}(0, 1)$.

Example 1.6.1 (cont)

1.6.3

The distribution of all order
statistics

PDF of all order statistics

For an i.i.d. sample of continuous random variables $X_1, X_2, \dots, X_n$ with pdf f , the pdf of all n order statistics is

$$\begin{aligned} & f_{X_{(1)}, X_{(2)}, \dots, X_{(n)}}(x_1, x_2, \dots, x_n) \\ &= n! f(x_1) f(x_2) \cdots f(x_n) I(x_1 < x_2 < \cdots < x_{n-1} < x_n). \end{aligned}$$

Proof

Proof

Proof

Proof

1.6.4

The distribution of $X_{(i)}$

CDF of $X_{(i)}$

For an i.i.d. sample of continuous random variables $X_1, X_2, \dots, X_n$ with cdf F , the cdf of $X_{(i)}$ is

$$F_{X_{(i)}}(x) = P(X_{(i)} \leq x) = \sum_{j=i}^n \binom{n}{j} (1 - F(x))^{n-j} F(x)^j.$$

Proof

Proof (cont)

The PDF of $X_{(i)}$

For an i.i.d. sample of continuous random variables $X_1, X_2, \dots, X_n$ with cdf F , the pdf of $X_{(i)}$ is

$$f_{X_{(i)}}(x) = \frac{n!}{(i-1)!(n-i)!} F(x)^{i-1} (1 - F(x))^{n-i} f(x).$$

Proof

Proof (cont)

Proof (cont)

1.6.5

The joint distribution of $X_{(i)}$ and $X_{(j)}$

The joint pdf of $X_{(i)}$ and $X_{(j)}$

For an i.i.d. sample of continuous random variables $X_1, X_2, \dots, X_n$ with cdf F , the joint pdf of $(X_{(i)}, X_{(j)})$ is

$$\begin{aligned} & f_{X_{(i)}, X_{(j)}}(x_i, x_j) \\ &= \frac{n!}{(i-1)!(j-i-1)!(n-j)!} F(x_i)^{i-1} f(x_i) \\ & \quad \times [F(x_j) - F(x_i)]^{j-i-1} f(x_j) [1 - F(x_j)]^{n-j} I_{(-\infty, x_j)}(x_i). \end{aligned}$$